

Donghee Hong

College Park, MD

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Research Interests

Visual analytics, human-AI interaction, large language models for visualization, human-in-the-loop model optimization, explainable AI

Education

University of Maryland, College Park (UMD)

PH.D. IN COMPUTER SCIENCE

College Park, MD, USA

Aug 2026 - Present

Sungkyunkwan University (SKKU)

M.S. IN ARTIFICIAL INTELLIGENCE

Suwon, Korea

Mar 2024 - Feb 2026

- Cumulative GPA: 4.5/4.5

Sungkyunkwan University

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Suwon, Korea

Mar 2020 - Feb 2024

- Cumulative GPA: 4.35/4.5 (98.5%), Major GPA: 4.38/4.5
- Graduated Summa Cum Laude, Class Rank: 2/160

Research Experience

Human-Data Interaction Research Group, UMD

RESEARCH ASSISTANT (ADVISER: PROF. LEO ZHICHENG LIU)

College Park, USA

July 2026 - Present

- Investigating how LLMs reason about visualization design choices and adapt to dataset properties and analytical goals
- Designing intermediate representations and interactive interfaces that connect visualization decisions, reasoning factors, and textual evidence

Interactive Data Computing Laboratory (IDCLab), SKKU

RESEARCH ASSISTANT (ADVISER: PROF. JAEMIN JO)

Suwon, Korea

Aug 2022 - Jun 2026

- Collaborated with domain experts in machine learning and software testing to characterize challenges in complex parameter optimization workflows
- Developed visual analytics methods for understanding parameter effects, comparing optimization runs, and exploring promising parameter configurations
- Led the design, implementation, and evaluation of **HyPockeTuner**, **Symetra**, and **TuneScape**, resulting in first-author publications at CHI, EuroVis, and IEEE VIS

Data-driven AI Security HCI (DASH) Laboratory, SKKU

RESEARCH ASSISTANT (ADVISER: PROF. SIMON SUNG-IL WOO)

Suwon, Korea

Dec 2020 - Jun 2022

- Contributed to computer vision research on deepfake detection and large-scale vision dataset construction

Publications

[1] TuneScape: Visualizing Multi-Tuner Parameter Optimization Logs

Donghee Hong, Minjong Kim, Sooyoung Cha, Jaemin Jo

IEEE Visualization and Visual Analytics Conference (VIS), Short Papers, 2026 [🌐 Code](#) [🖥️ Demo](#)

[2] Symetra: Visual Analytics for the Parameter Tuning Process of Symbolic Execution Engines

Donghee Hong, Minjong Kim, Sooyoung Cha, Jaemin Jo

Eurographics Conference on Visualization (EuroVis), 2026 [📄 Paper](#) [🌐 Code](#) [🖥️ Demo](#)

[3] HyPockeTuner: Bringing Hyperparameter Optimization to Mobile Devices

Donghee Hong, Bongshin Lee, Jinwook Seo, Jaemin Jo

ACM Conference on Human Factors in Computing Systems (CHI), 2026 [📄 Paper](#) [🌐 Code](#)

[4] **Symetra: Visual Analytics for Tuning Symbolic Execution Engines**
Donghee Hong, Minjong Kim, Sooyoung Cha, Jaemin Jo
IEEE Visualization and Visual Analytics Conference (VIS), Poster, 2025

[5] **LLM-Based Text-to-Image Prompt Conversion System for Custom Wheelchair Spoke Guard Design**
Seongwan Park, Jiyeol Park, **Donghee Hong**
Korea Computer Congress (KCC), 2024  Paper

[6] **Deepfake Detection for Fake Images with Facemasks**
Sangjun Lee, Donggeun Ko, Jinyong Park, Saebyeol Shin, **Donghee Hong**, Simon S. Woo
Workshop on Security Implications of Deepfakes and Cheapfakes (WDC), 2022  Paper

[7] **VFP290K: A Large-Scale Benchmark Dataset for Vision-Based Fallen Person Detection**
Jaeju An, Jeongho Kim, Hanbeen Lee, Jinbeom Kim, Junhyung Kang, Minha Kim, Saebyeol Shin, Minha Kim, **Donghee Hong**, Simon S. Woo
Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2021  Paper  Code

Selected Projects

PediaViz: Interactive Timeline Dashboard for Pediatric Medication and Test Analysis
IDCLAB, SKKU COLLABORATING WITH SEOUL NATIONAL UNIVERSITY HOSPITAL & NAVER CORPORATION
Feb 2024 - Aug 2026

- Designed and implemented a web-based dashboard that aligns patient growth records, lab results, and medication history on a shared timeline
- Extended the system to pair structured clinical measurements with unstructured diagnostic text, so that numeric trends and physician assessments can be read together
- Built custom visualization components for comparing treatment history against test results over time, developed through iterative feedback from pediatricians

Spokeguard Design Assistant for Young Wheelchair Users
CO-DEEP LEARNING PROJECT, SKKU
Sep 2023 - Dec 2023

- Built an LLM-based system that converts Korean design descriptions into text-to-image prompts, allowing children to specify wheelchair spoke guard designs without writing English prompts (KCC 2024)

Selected Honors & Awards

Dean's Fellowship, UMD
Competitive fellowship for outstanding incoming graduate students
2026 – 2028

Honorable Mention, Graduate Student Paper Award, SKKU
2025

Graduate Scholarship, SBS CULTURAL FOUNDATION
15 recipients nationwide, 4% acceptance rate
2024 – 2026

Grand Prize, Co-Deep Learning Project Competition, SKKU
Top 2 of 16 teams
2023

Presidential Science Scholarship, KOREA STUDENT AID FOUNDATION
National scholarship awarded to 60 outstanding third-year STEM students
2022 – 2023

Technical Skills

Programming	Python, C/C++, JavaScript/TypeScript, Java
Machine Learning & AI	PyTorch, Scikit-learn, Pandas, NumPy
Generative AI	OpenAI API, Claude API, Hugging Face Transformers, Stable Diffusion
Visualization & Web	D3.js, visx, React, Tailwind CSS, HTML/CSS, FastAPI
Research Methods	Formative Studies, Controlled Lab Studies, Case Studies, Semi-structured Interviews, Thematic Analysis
Tools	Git, LaTeX, Figma